Capstone Project 1: Data Collection

Course Title: Introduction to Data and Knowledge and Ontology

Assignment Type: Individual
Submission Format: Report(s) + Dataset File(s)
Due Date: 19rd April 2026 23:59

Objective:

The goal of this project is to apply knowledge of data collection techniques to gather, clean, and organize data relevant to one or more specific case study(s). By the end of this project, students will have experience in sourcing real-world datasets, designing a data collection strategy, and performing initial data cleaning.

Project Guidelines:

Step 1: Selection of Case Study (Research & Justification)

- Choose two case studies that interests you. The case study should focus on a specific problem or question that can be analyzed using collected data.
- Example topics:
    - Sentiment analysis on social media data
    - Customer purchase behavior analysis
    - Stock market trends analysis
    - Healthcare data trends
    - Environmental data patterns
- **Deliverable:** A 1-2 page document outlining your case study(s), its importance, and how data collection can help answer key questions.

Step 2: Identifying the Source of the Data

- Determine the type of data:
    - Structured vs. Unstructured Data
    - Quantitative vs. Qualitative Data
- Identify potential data sources:
    - Public datasets (e.g., Kaggle, Google Dataset Search, UCI ML Repository)
    - Private datasets (e.g. the company you are interning at)
    - APIs (Twitter API, OpenWeather API, Financial APIs)
    - Web scraping (News articles, e-commerce prices, job postings)
    - Surveys and First-hand data collection
- **Deliverable:**
    - A document listing at least **three** potential data sources
    - and why they are relevant.

Step 3: Devising a Data Collection Strategy

- Define:
  - The method of collection (**API, Web Scraping, Surveys, Public Repositories, Manual Entry**)
  - Frequency of data collection (Real-time, Batch, One-time retrieval)
  - Tools and technologies needed (**Python, BeautifulSoup, Scrapy, Pandas, SQL, Postman, etc.**)
  - If the data was sent to you from someone, describe the process steps, tools and technologies they used to extract the data. If the data comes from another student, this could be considered as cheating. This is only applicable to students' collecting data from their intern companies. If you don't have this information, you will not get the marks for this section as it doesn't show your understanding on data collection. Same goes for pre-processing.
  - Storage format (**CSV, JSON, SQL Database, NoSQL Database**)
- **Deliverable:**
  - A well-documented plan describing your data collection strategy
  - Tools used,
  - and how the collected data will be stored.

Step 4: Collecting the Data

- Implement **2** of the data collection strategy for each of the 2 datasets.
- Download, extract, or scrape the data.
- Store the data in a structured format.
- Ensure completeness by checking for missing records.
- Make sure the dataset is large enough with at least 10 000 records or 100 000 words if unstructured.
- **Deliverable:**
  - Short report on how the data was collected (for those that have code, just the code is sufficient). Not more than 1 page description.
  - Uncleaned dataset in an appropriate format (CSV, JSON, or SQL dump),

Step 5: Cleaning the Data

- Perform initial cleaning steps:
  - Handling missing data (imputation, deletion, etc.)
  - Removing duplicate records
  - Data normalization (if applicable)
  - Handling outliers (basic anomaly detection)
  - Ensuring consistent formats (date, text, numbers)
- The result of the collected data should have at least **15** data points (attributes) across all collections.
- For the above bullet points, if your data doesn't require the steps provided, theoretically justify why your dataset did not require it. For example:

    - *“The data set didn’t require the handling of missing data because out of the x amount of records and y amount of columns, there wasn’t any missing values found”*
    - *“The dataset didn’t require outlier removal since the normal distribution before outlier removal before looks like {this} and after outlier removal looks like {this} with only x amount of different is mean” any other statistical justification that makes sense can be considered here.*
- **Deliverable:**
    - A report justifying the data cleaning techniques applied
    - Cleaned dataset with the described format of each column.

Submission Requirements

Your final submission should include:

1. **Case Study Selection Report (Step 1)**
2. **Data Source Identification (Step 2)**
3. **Data Collection Strategy Document (Step 3)**
4. **Dataset File (Step 4)**
5. **Data Cleaning Report (Step 5)**

Evaluation Criteria

Category	Points	Description
Case Study Relevance	10	Clarity and significance of the chosen case studies (2 in total).
Data Source Selection	15	Justification and variety of data sources (2 per case study).
Data Collection Strategy	20	Well-structured plan using appropriate methods and tools.
Dataset(s) Completeness	20	Quality and comprehensiveness of collected data. At least 15 data points (attributes) Across all data points (2 datasets).
Data Cleaning	20	Effectiveness in handling missing values, duplicates, and formatting.
Documentation & Clarity	15	Well-explained processes, readable submission.

Total: 100 points